

AI concepts for official statisticians

Mechanism, evidence and accountable decisions



Learning outcomes and professional boundary

- Decompose statistical work and locate retained judgement
- Distinguish mechanisms, assumptions and uncertainty
- Trace failures into official products and institutional duties
- Issue a documented permit, redesign, reject or escalate decision



Quality and AI assurance

- Evaluate quality at the official-product level
- Treat confidentiality, validity and authority as mandatory gates
- Compare quality-adjusted time with the current baseline
- Record provenance, versions, corrections and accountable owners



Definitions and system boundaries

- Describe observable system behaviour without assigning human faculties
- Separate encoded rules, estimated parameters, retrieval, generation and action
- Use the current method as the baseline for end-to-end comparison



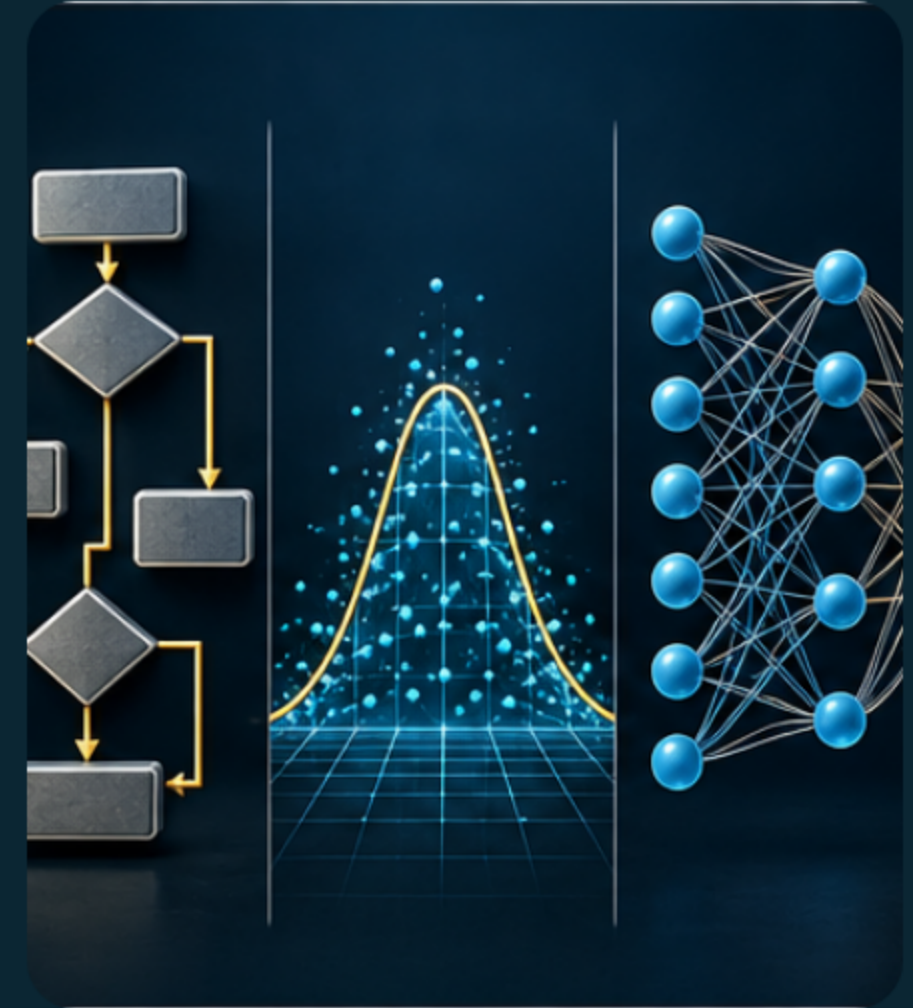
Statistical purpose and task decomposition

- Begin with the user, population, concept, reference period and consequence of error
- Map tasks as functions with inputs, outputs, owners and acceptance conditions
- Subject-matter statisticians retain construct, evidence and release boundaries



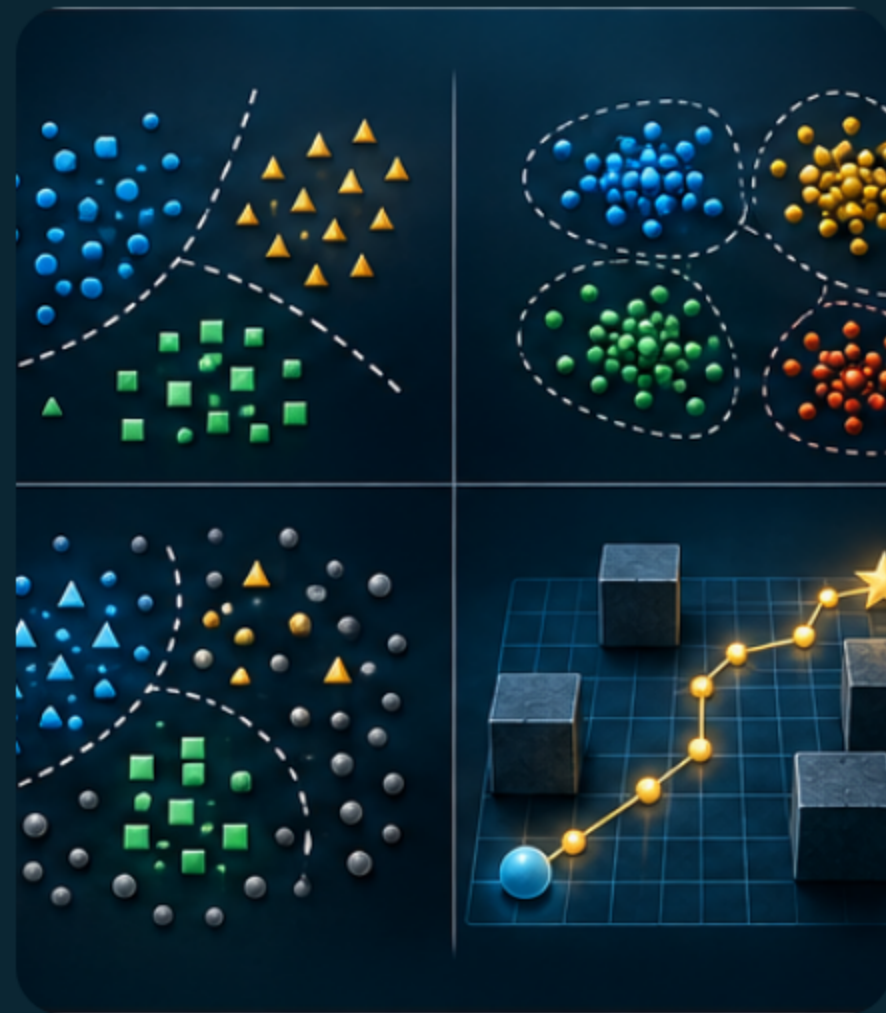
Rules, models and machine learning

- Rules encode conditions, models express stochastic relationships and learning estimates functions
- Specification validity and empirical behaviour require different evidence
- Fitness depends on population, error consequences and integration into estimation



Learning families and their evidence

- Labels, clusters, pseudo-labels and rewards create different validity questions
- Loss and stability depend on sampling, weighting, interpretation and error costs
- Exploratory discovery must remain separate from consequential production action



Deep representation learning

- Layered representations can capture complexity while suppressing rare distinctions
- Test perturbation sensitivity, domain slices, run-to-run stability and simpler baselines
- Technical implementation remains bounded by the statistical construct and mandate



Generative and foundation models

- One model can draft, classify, translate or code, but each use needs separate evidence
- Plausible token generation does not establish factual or statistical validity
- Prompting controls an input but cannot replace independent validation



Retrieval, grounding and citations

- Trace document selection, ranking, context assembly and generation separately
- Test corpus coverage, recall, faithful use and claim-level support
- Refusal and referral are valid outputs when evidence is insufficient



Optimisation, recommendation and agents

- Moving from prediction to action increases consequence and feedback risk
- Objectives cannot absorb legal, ethical and statistical hard constraints
- Tool permissions, logs, transaction limits and stopping rules bound agentic work



Training, evaluation and operational use

- Training fit, independent evaluation and production inference support different claims
- Declare population, reference standard, metrics, uncertainty and thresholds
- Good fit does not establish operational feasibility or authority to use



Generalisation and distribution drift

- Input, prevalence and concept relationships can all shift after development
- Production loss can differ from development loss even when code is unchanged
- Monitoring combines data diagnostics, reference samples and institutional events



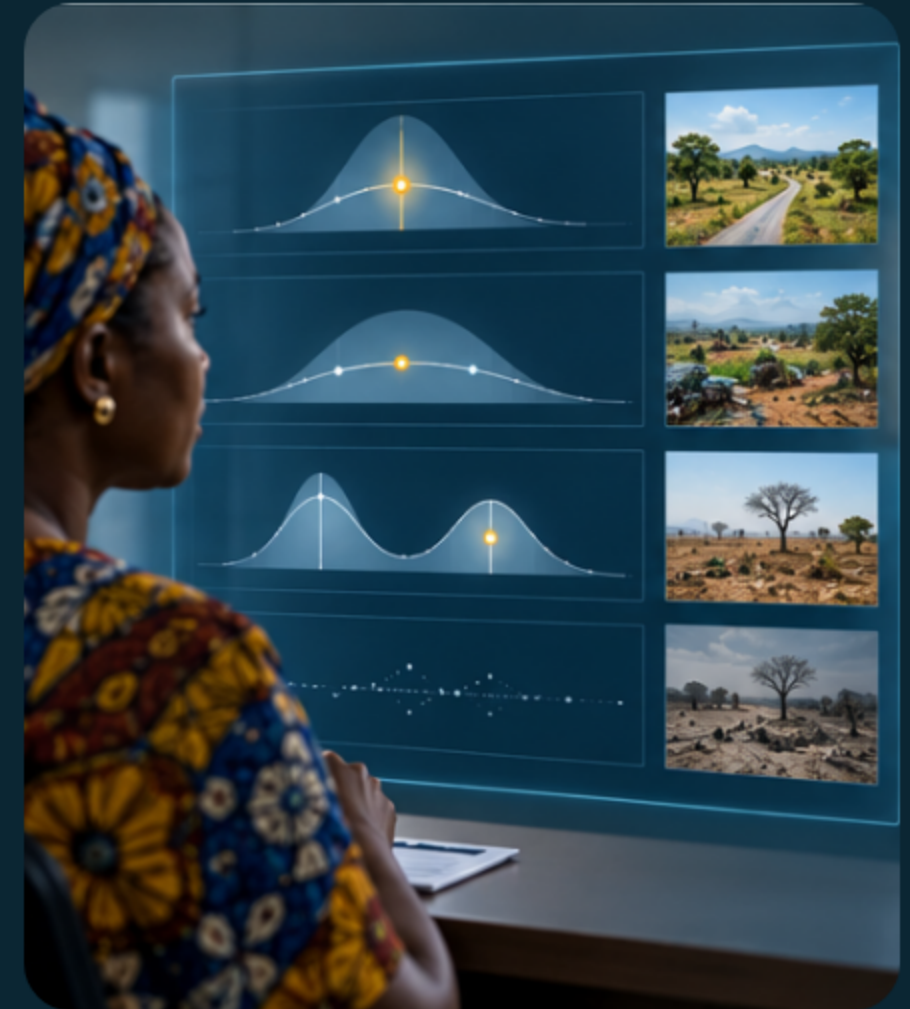
Leakage, coverage, selection and proxies

- Leakage, population coverage, selection and proxy validity are distinct defects
- Evaluation splits must respect entity, time, household and geographic structure
- A precise predictor over a distorted frame can worsen an official estimate



Hallucination, calibration and uncertainty

- Separate unsupported claims from predictive, sampling and measurement uncertainty
- Calibration requires empirical evidence under relevant conditions
- Retain an uncertainty ledger instead of collapsing evidence into one score



Subgroup performance and public value

- Aggregate performance can conceal failure for languages, regions or rare groups
- Fairness metrics can conflict and subgroup estimates carry uncertainty
- Mandate, consequence, participation and appeal guide the value choice



Automation bias and institutional memory

- Oversight fails when reviewers lack time, independence, information or authority
- Selective review can miss confident systematic errors and amplify feedback loops
- Fallback, blind samples and recovery exercises preserve institutional capability



Confidentiality, cybersecurity and suppliers

- Prompts, embeddings, logs, artefacts and telemetry create new protected data flows
- Threat models cover access, injection, extraction, poisoning and supply-chain change
- Procurement cannot transfer the institution's legal or professional responsibility



Reproducibility and auditability

- Controlled inputs, versions and environments must regenerate declared outputs
- Evidence records retain lineage, configuration, results, exceptions and approvals
- Independent reviewers must reconstruct what happened and why it was allowed



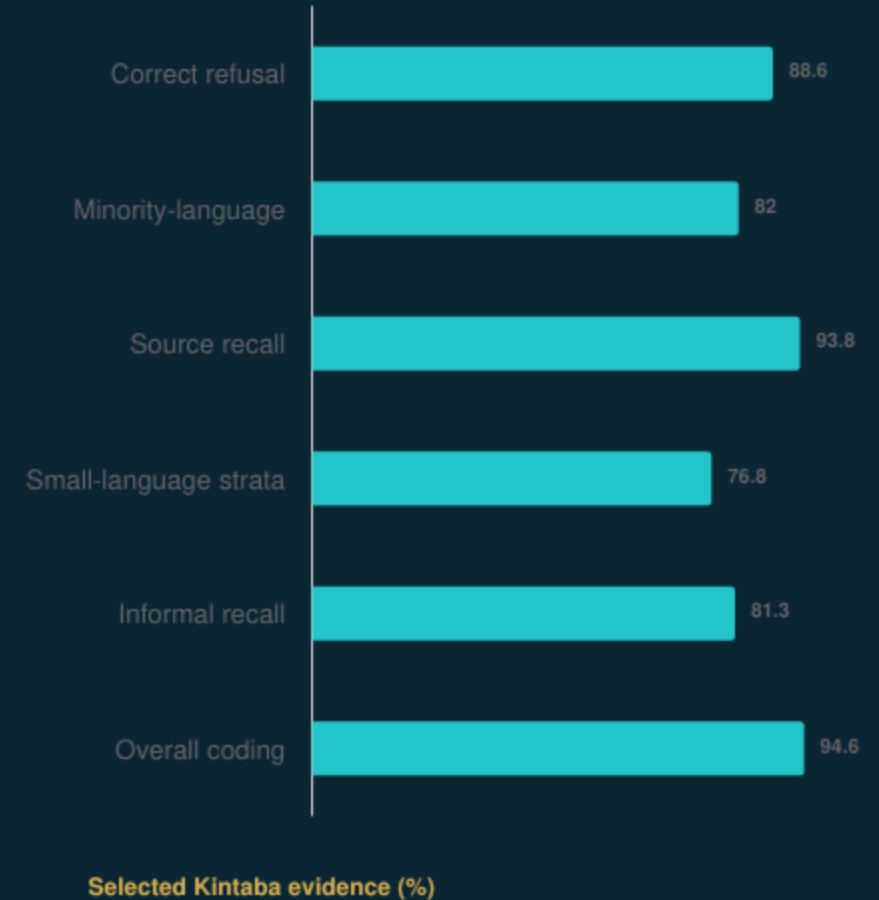
Human oversight and stopping rules

- Meaningful oversight requires competence, information, time and power to intervene
- Stopping rules name the signal, threshold, owner, containment and restart conditions
- Systems may propose or draft, but cannot acquire statistical or release authority



Kintaba cases across the value chain

- Cases compare coding, linkage, crop estimation, drafting and public query support
- Each dossier records evidence, tolerances, uncertainty, controls and owner
- Strong aggregate results cannot compensate for a failed mandatory gate



Technology selection and proportional assurance

- Compare manual, rule-based, statistical, machine-learning and hybrid alternatives
- Assess validity, total effort, subgroup results, reversibility and supplier dependence
- Increase assurance with consequence, uncertainty, irreversibility and scale



Decision record and workplace transfer

- Record purpose, baseline, evidence, alternatives, conditions and residual uncertainty
- Localise the decision to mandate, law, language, infrastructure and workforce
- Begin with an authorised bounded task and preserve the record through staff change

